



ECE 501: Digital Image Processing

Week 1 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

<https://github.com/ERYAGNIK003/DRASHTI-Dataset>

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

<https://zenodo.org/records/18278989>

1. Work Done This Week

project topic named: Vehicle Orientation Estimation Using Conventional Image Processing and began to understand the problem statement, dataset and expected results.

We examined DRASHTI-HAORB to gain insight into the DRASHTI-HAORB annotation that provides vehicle information. The oriented bounding box (OBB) coordinates, vehicle class, difficulty level and heading angle were studied for each vehicle. The heading angle is given in the range of 0° – 359° and is defined as the angle, measured clockwise from the x-axis of the image coordinate system, of the front of the vehicle. This heading angle also had to be converted to the 0° – 90° orientation range expected for the project, which we explored during the initial week.

We explored the structure of the dataset, as well as the image and annotation files. We also learned about the annotation format: the coordinates of the four corners of the OBB followed by the class label, the level of the difficulty, and the heading angle.

We developed and organized a public GitHub repository, to keep all project files, project code, results and progress for the entire project. We also began to look at the files of the available data set and the visualization tools to see how the data set are correlated to the images of the vehicles.

Along with the dataset study, we started a preliminary literature review on conventional image-processing methods that can be used for estimating the orientation of an object. We studied methods based on contour, Principal Component Analysis (PCA), geometric features and line/edge detection methods like the Hough Transform. At this point, we haven't yet finalized any of the methods, and are still thinking about which one might be most appropriate for the various categories of vehicle images in the dataset.

We also spoke about how the project would be accomplished overall. The proposed procedure is to extract vehicle crops from the OBB annotations, identify the orientation of the image based on classical image-processing techniques, and make a comparison between the estimated orientation and the ground-truth orientation.

The major stages identified for the project are:

- Extract vehicle crops using OBB annotations.
- Convert heading angles to the required 0° – 90° range.
- Apply conventional image-processing methods for orientation estimation.
- Compare estimated and ground-truth orientations.
- Calculate angular error and evaluate performance.
- Analyze the effect of vehicle size, quality, and orientation.

2. Literature Review

A preliminary survey of the literature was conducted to gain knowledge of the vehicle orientation estimation and the conventional image-processing methods.

We conducted a literature survey of the work done on vehicle orientation estimation and considered PCA, contour-based, geometric feature, and Hough Transform approaches. The dominant direction of a vehicle shape can be determined by looking for the dominant direction in PCA, and the orientation can be estimated by using the silhouette or the bounding shape of the vehicle with contour or geometric methods. Prominent lines can be found by the edge information using Hough Transform.

These methods were taken into consideration due to their relevance in our dataset and project requirements. So we are not in the final stages on how to do this, and will experiment with various methods in the coming weeks to see which ones work best.

3. Work to Be Done Next Week

In Week 2, we will be focusing on the practical aspect of the project and gain further insight into our data:

- Download the necessary DRASHTI-HaOBB images and original annotations.
- Programmatically check annotation files and verify OBB coordinates, class labels, difficulty and heading angles.
- Create a preliminary approach to separating the individual car crops from the OBB annotations.
- Apply the conversion of 0° – 359° heading angle to the representation of the required 0° – 90° orientation on the sample images to verify it.
- Create and visually evaluate a small number of vehicle crops and ground truth orientations.
- Start to apply and experiment with basic conventional image-processing techniques, specifically PCA/contour-based and geometric.
- Investigate if Hough-transform based orientation estimates are helpful.
- Define appropriate evaluation methods for the comparison of estimated and ground-truth orientations and highlight the comparison results of initial tests.

4. Challenges Faced

This was the first week and the primary difficulties were with understanding the dataset and determining the approach to the orientation estimation problem.

The first difficulty was deciding how the 0° to 359° (compass direction) heading angles that were available in the data should be presented as 0° to 90° . This has to be done in a uniform manner.

Another observation was that the size, shape, orientation and image quality of the vehicles varies, which could impact the effectiveness of conventional image processing techniques. Vehicle images may also be somewhat cropped or blurry.

Another difficulty is the choice of the method that will provide the most accurate results. Consideration is being given to PCA, contour/geometric methods and Hough Transform, but more testing of the data set is needed to choose between the methods.

5. Current Status

Project has been set up and initial dataset study completed at the end of Week 1. We have recognized the key points that are needed for the project and begun investigating some of the conventional image-processing methods that could be used.

The orientation-estimation method has not been determined as yet. In the next phase, we will work to apply and evaluate the preliminary strategies with sample images of vehicles and learn how they perform.

6. References

- [1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, “Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB),” *IEEE Data Descriptions*, 2026, doi: 10.1109/IEEEDATA.2026.3670752.
- [2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, “DRASHTI-HaOBB: Drone nadir-view Annotated imageS of veHicles dataset for India - Heading-angle Oriented Bounding Box,” *Zenodo*, 2026, doi: 10.5281/zenodo.18278989.
- [3] “Learning single-shot vehicle orientation estimation from large-scale street panoramas,” *Neurocomputing*, vol. 367, pp. 319–327, 2019.
- [4] “Vehicle Position Monitoring Using Hough Transform,” *IERI Procedia*, vol. 4, pp. 316–322, 2013.
- [5] “Detection of vehicles from traffic scenes using fuzzy integrals,” *Pattern Recognition*, vol. 35, no. 4, pp. 967–980, 2002.